Natural Language PDDL (NL-PDDL) for Open-world Goal-oriented Commonsense Regression Planning in Embodied AI

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Motivation

Problem:

Open-world plans must be both **sound** and support **expressive** domains with 1) **long horizon** causal dependencies, 2) unstructured **visual and language** observation and 3) **incomplete** domain knowledge.

- Symbolic Planner ⟹ sound, but **not expressive**
- LLM Planner ⟹ struggles **with long-horizon soundness**
- Both approaches ⟹ struggle with **incomplete domain knowledge**

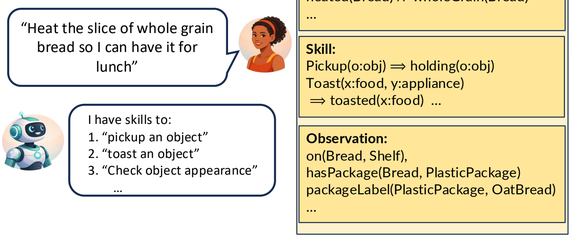
Solution:

NL-PDDL is a goal-oriented regression framework that integrates predicate-level natural language representation with LLM-based commonsense for open-world task planning.

Key Features

1. **NL Predicates:** Use natural language to represent predicates, enabling formal modeling of states, actions, and goals in NL.
2. **Commonsense Entailment:** Use LLM for commonsense reasoning to infer entailment relations among NL-predicates.
3. **NL Goal-Regression:** Generate structured NL plans and subgoals via goal-regression to guide sensing with LLMs and VLMs.

Natural Language(NL) Predicates



Symbolic (PDDL)

Goal:
heated(Bread) ∧ wholeGrain(Bread)
...

Skill:
Pickup(o:obj) ⟹ holding(o:obj)
Toast(x:food, y:appliance)
⟹ toasted(x:food) ...

Observation:
on(Bread, Shelf),
hasPackage(Bread, PlasticPackage)
packageLabel(PlasticPackage, OatBread)
...

NL-Predicate (NL-PDDL)

Goal:
"(Bread} is heated", and
"(Bread) is whole grain", and
...

Skill:
"If I pick up (x:obj)" ⟹ "I am holding (x:obj)"
"If I toast (x:obj) with (y:appliance)"
⟹ "(x:obj) is toasted" ...

Observation:
"I see a slice of (Bread) on the shelf."
"(Bread) has the word 'Oat Bread' on the plastic package" ...

Expressive Representation:
NL-predicates are more expressive than rigid symbols, enabling commonsense reasoning among them.

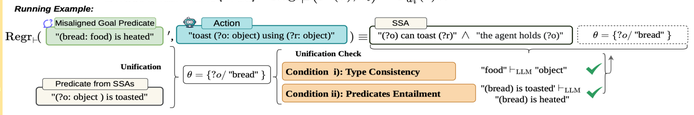
"(Bread) has label 'Oat Bread' on package" ⊨ "(Bread) is whole grain"
"(Bread) is Whole Wheat " ⊨ "(Bread) is whole grain"

Misaligned Goals and Actions:
Using NL-Predicates, NL-PDDL can plan despite semantic misalignment between goals and action models.

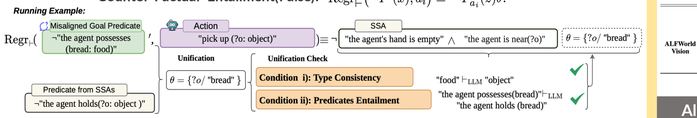
"(Bread) is toasted" ⊨ " (Bread} is heated"
"(Bread) is microwaved" ⊨ " (Bread} is heated"

Natural Language Entailment

Direct Entailment (True): $\mathrm{Regr}_{\vdash}(P'(\vec{x}), a_i) \equiv F'_{a_i}(z)\theta.$

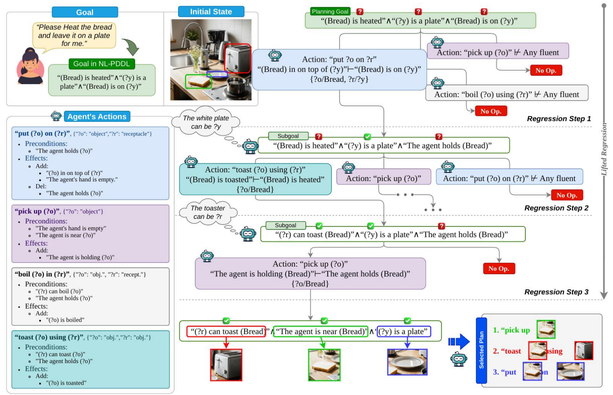


Counter-Factual Entailment(False): $\mathrm{Regr}_{\vdash}(\neg P'(\vec{x}), a_i) \equiv \neg F'_{a_i}(\vec{z})\theta.$



Can we perform Formal Reasoning directly with Natural Language to enable sound and expressive planning in the Open-World? Yes! With NL-PDDL

NL-PDDL Goal Regression



NL-PDDL Goal Regression Components

Effect Axioms: Determine causal relation among action and facts

Running Example (continued):

IF "I am holding object x"
AND "object y can toast object x"
AND "I toasted object x with y"
THEN "object x is toasted after I acted"

Positive effect axiom for Predicate "(?o : object) is toasted" with respect to the Action "toast (?o : object) using (?r : object)"

$a = \langle$ Action "toast (?o) using (?r)" , Precondition "(?r) can toast (?o)" ∧ "the agent holds (?o)" $\rangle \Rightarrow$ Next-state Predicate "(?o) is toasted"

$\gamma^{+}_{\text{Pleasant}, \theta_{\text{toast}}}(?o, ?r)$

Negative effect axiom for Predicate "the agent holds (?o : object)" with respect to the Action "put (?o : object) on (?r : receptacle)"

$a = \langle$ Action "toast (?o) using (?r)" , Precondition "the agent is near (?r)" ∧ "the agent holds (?o)" $\rangle \Rightarrow$ Next-state Predicate "¬the agent holds (?o)"

$\gamma^{-}_{\text{Pleasant}, \theta_{\text{put}}}(?o, ?r)$

$\forall \vec{y} : \vec{T}\left[\gamma^{+}_{F,a_i}(\vec{y}) \Rightarrow F'_{a_i}(\vec{y_F})\right], \quad \forall \vec{y} : \vec{T}\left[\gamma^{-}_{F,a_i}(\vec{y}) \Rightarrow \neg F'_{a_i}(\vec{y_F})\right].$

Successor State Axioms: What facts hold after an action

Running Example (continued):

"object x is toasted after I acted"
IFF "I am holding object x"
AND "object y can toast object x"
AND "I toasted object x with y"
OR "object x is already toasted before I acted"

Successor state axiom for Predicate "(?o : object) is toasted" and Action "toast (?o : object) using (?r : object)"

Next-state Predicate "(?o) is toasted" = Predicate "(?o) is toasted" ∨ ($a =$ Action "toast (?o) using (?r)" ∧ Precondition "(?r) can toast (?o)" ∧ "the agent is near (?r)")

$\gamma^{+}_{\text{Pleasant}, \theta_{\text{toast}}}(?o, ?r)$

Successor state axiom for Predicate "the agent holds (?o : object)" and Action "put (?o : object) on (?r : receptacle)"

Next-state Predicate "the agent holds (?o)" = Predicate "the agent holds (?o)" ∧ ¬($a =$ Action "put (?o) on (?r)" ∧ Precondition "the agent is near (?r)" ∧ "the agent holds (?o)")

$\gamma^{-}_{\text{Pleasant}, \theta_{\text{put}}}(?o, ?r)$

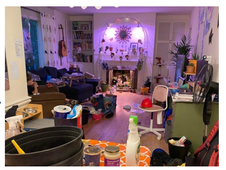
$F'_{a_i}(\vec{x}) \equiv \gamma^{+}_{F,a_i}(\vec{x}) \vee \left(F(\vec{x}) \wedge \neg\gamma^{-}_{F,a_i}(\vec{x})\right).$

Guided Sensing (Subgoal/Plan)

Exhaustive Enumeration
" *Give me all facts you can observe*"
⟹ Combinatorial explosion

Goal Based Sensing
" *Give me all facts related to the goal*"
⟹ Ambiguity and missing information

Subgoal/Plan Based Sensing
" *Give me all facts related to the plan*"
⟹ Plan contains all the information to achieve the goal



Experiments and Evaluation



Open-World Without Misalignment

Datasets	Methods		# Tok.	# Expert Trajectory	Fine-Tuning	SR (%)
ALFWorld Text	Direct LLM-based Planner	GPT-4o	1,358,934	0	✗	21%
		Gemini-2.0-Flash	1,176,406	0	✗	16%
		LLaMA-3.1	1,193,640	0	✗	19%
	Reflective LLM-based Planner	ReAct (w examples)	5,507,266	0	✗	81%
		ReAct (w model)	4,242,560	0	✗	34%
		Reflexion-3[†]	NA	0	✗	83%
		Reflexion-10[†]	NA	0	✗	91%
		DEPS[†]	NA	0	✗	76%
	Fine-tuned LM	BUTLER	NA	100,000	✓	26%
	Ours	NL-PDDL	443,407	0	✗	94%
ALFWorld Vision	Direct VLM-based Planner	GPT-4o	1,823,539	0	✗	8%
		Gemini-2.0-Flash	1,640,025	0	✗	2%
		LLaMA-3.1	1,505,371	0	✗	2%
	Fine-tuned VLM	LLaMA-Adapter[†]	NA	170,000	✓	13%
		InstructBLIP[†]	NA	170,000	✓	22%
		EMMA-3[†]	NA	15,237	✓	37%
		EMMA-10[†]	NA	15,237	✓	82%
	Ours	NL-PDDL	679,148	0	✗	84%

Closed-World Without Misalignment

Method	Blockworld # Tok.	Blockworld SR (%)	Mystery Blockworld # Tok.	Mystery Blockworld SR (%)	Randomized Blockworld # Tok.	Randomized Blockworld SR (%)
GPT-4o	982,602	34%	834,890	0%	835,974	0%
Gemini-2.0 Flash	928,024	18%	834,060	1%	835,824	0%
LLaMA 3.1	963,565	44%	841,791	0%	842,622	0%
Fast Downward	N/A	100%	N/A	100%	N/A	100%
NL-PDDL	0	70%	0	70%	0	70%

Open-World: High success and efficiency, outperforming LLM/VLM baselines across text and visual modalities.

Closed-World: Strong performance in classical domains, robust to domain shifts (Mystery/Randomized).

Aligned Domain

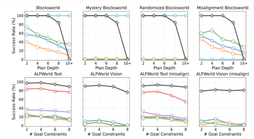
Misaligned Benchmarks

Category	Method	ALFWorld Text # Tok.	ALFWorld Text SR (%)	ALFWorld Vision # Tok.	ALFWorld Vision SR (%)	Blockworld # Tok.	Blockworld SR (%)
Direct LLM	GPT-4o	1,440,470	17% (↓3%)	1,811,404	7% (↓1%)	937,905	27% (↓7%)
	Gemini-2.0 Flash	1,251,533	15% (↓1%)	1,202,016	5% (↑3%)	939,050	21% (↑3%)
	LLaMA-3.1	1,310,664	15% (↓4%)	1,480,392	2% (↓0%)	950,490	42% (↓3%)
ReAct	w/ examples	5,213,612	79% (↓1%)	N/A	N/A	N/A	N/A
	w/ model	4,428,914	23% (↓11%)	N/A	N/A	N/A	N/A
Classical Planner	Fast Downward	N/A	N/A	N/A	N/A	0	0%
Ours	NL-PDDL	501,049	**91%** (↓2%)	745,505	**80%** (↓4%)	11,656	**70%**

Open-World Misalignment: Maintains high success while LLM planners degrade significantly.

Closed-World Misalignment: Remains robust under goal–action mismatch, unlike classical planners that fail without matched symbols

Misaligned Domain



Task Complexity: Remains stable as goal complexity increases, outperforming LLM/VLM planners that degrade sharply.

Plan Horizon: Strong performance at moderate depths, with graceful degradation at longer horizons due to runtime limits.

Scalability

Conclusion and Insights

1. **Natural Language Representation:** Enables LLM based commonsense reasoning over rich multimodal domains without requiring manually defined axioms.
2. **Subgoal Guided Sensing:** Uses plan-inferred subgoals to guide LLM/VLM queries, avoiding combinatorial explosion and relevance under-specification.
3. **Goal-Regression Reasoning:** Provides a mechanism for Open-world (lifted) plan generation and plan-guided sensing without complete domain knowledge.

Future Direction

- LLM/VLM based model generation & refinement
- Planning integration with LLM agents with skill & memory
- RL Integration for exploration guidance via regression reasoning

Paper Code

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